Chain-of-Thought Prompting in Large Language Models: Mechanisms, Performance Gains, and Limitations of Reasoning

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Abstract

Chain-of-thought (CoT) prompting has become an effective technique for improving reasoning performance in large language models (LLMs), particularly on tasks requiring multi-step computation and logical inference. Yet the underlying reasons for these improvements remain not fully understood. This paper investigates how CoT prompting works and why it leads to better performance, focusing on both its mechanisms and its limitations. We analyze CoT from three complementary perspectives. First, we examine how generating intermediate reasoning steps enables models to decompose complex problems into simpler subproblems, effectively restructuring the inference process. Second, we review existing findings showing that CoT significantly improves accuracy on reasoning benchmarks, especially at larger model scales and on tasks requiring sequential decision-making. Third, we explore the limitations of CoT, including sensitivity to prompt design, systematic reasoning errors, and the lack of faithfulness between generated explanations and actual model behavior. Our analysis suggests that CoT improves performance by transforming reasoning into a sequence of structured, intermediate predictions, allowing models to allocate computation across multiple steps rather than a single output. This leads to more accurate results on complex tasks, but does not guarantee that the underlying reasoning is correct or faithful. As a result, improving the reliability and consistency of these reasoning processes remains a key challenge, pointing toward future work on structured prompting, verification, and reasoning-aware training methods.

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1 Introduction

Recent progress in natural language processing (NLP) has been largely driven by transformer-based large language models (LLMs), which have significantly changed how many language tasks are approached. As these models grow in size, their performance continues to improve across a wide range of tasks. Scaling has been shown to provide clear benefits, including better accuracy and improved sample efficiency(Wei et al., 2022). Beyond these improvements, larger models also begin to show new abilities, such as solving tasks through in-context learning: instead of requiring additional training, LLMs can adapt to new problems by using a few examples or simple instructions provided in the prompt, allowing them to generalize effectively across different tasks (Wang et al., 2023a). Because of this, prompting has become a key way to interact with modern language models and plays an important role in recent advances in reasoning.

Despite these advances, reasoning remains a significant challenge for LLMs. While larger models exhibit improved performance overall, scaling alone does not guarantee strong reasoning abilities, particularly for tasks that require multi-step computation or logical inference (Wei et al., 2022). Even when models are able to generate answers, they often produce incomplete or incorrect reasoning processes, including missing intermediate steps, calculation errors, and semantic misunderstandings (Wang et al., 2023a). These limitations highlight a key gap between general language understanding and reliable reasoning, motivating the need for methods that can better structure and guide the reasoning process in LLMs.

Chain-of-thought prompting has been proposed as a simple yet effective method for improving reasoning in LLMs. CoT prompting encourages models to generate intermediate natural language reasoning steps before producing a final answer, effectively transforming the task from direct prediction into step-by-step problem solving (Wei et al., 2022). By structuring outputs as sequences of reasoning steps, CoT allows models to decompose

complex problems into smaller, more manageable components. This approach has been shown to significantly improve performance on a variety of reasoning benchmarks, particularly for large-scale models. Recent work further shows that these benefits extend beyond text-based models, with chain-of-thought reasoning improving performance and interpret-ability in multi-modal systems such as vision-language models (Zhang et al., 2025).

Nevertheless, the reason why CoT improves performance remains an open question. In particular, it is unclear whether CoT enables genuine reasoning or simply produces more structured outputs that approximate reasoning behavior. This paper investigates the following research question: How does chain-of-thought reasoning influence the performance of large language models, and why does it lead to improved results?

To address this question, this paper analyzes CoT prompting from three complementary perspectives. First, we examine the mechanisms that enable CoT to improve reasoning by decomposing complex tasks and structuring intermediate steps. Second, we evaluate the performance gains associated with CoT across reasoning benchmarks, highlighting how structured reasoning improves accuracy. Third, we analyze the limitations of CoT, including reasoning errors, bias, and lack of faithfulness, and discuss recent approaches such as Plan-and-Solve prompting that aim to improve reasoning reliability (Wang et al., 2023a). Together, these analyses provide a comprehensive view of both the strengths and limitations of CoT reasoning in modern language models.

2 Background

CoT prompting refers to a technique in which a language model is encouraged to generate intermediate reasoning steps before producing a final answer. Instead of directly mapping an input to an output, the model produces a sequence of natural language steps that reflect a reasoning process leading to the solution. As described by (Wei et al., 2022), a chain of thought can be understood as a series of intermediate reasoning steps that lead to the final output. These steps resemble how humans approach complex problems, where the solution is constructed gradually rather than produced immediately. By generating these intermediate explanations, the model is able to break down complex tasks into smaller components, making reasoning tasks more manageable.

A key distinction between standard prompting and CoT prompting lies in how the model approaches the problem. In standard prompting, the model is given input–output examples and is expected to produce an answer directly, without explicitly showing its reasoning. In contrast, CoT prompting augments these examples with step-by-step explanations, guiding the model to follow a structured reasoning process. This difference becomes especially important for tasks that require multiple steps, such as arithmetic or logical reasoning. For example, a standard prompt may produce only a final answer, while a CoT prompt generates intermediate calculations before arriving at that answer. Empirical evidence shows that this structured approach enables models to solve problems that they would otherwise fail under standard prompting, particularly on multi-step reasoning tasks (Wei et al., 2022).

The effectiveness of CoT prompting can also be better understood in the broader context of intermediate reasoning supervision. Prior work in reading comprehension shows that training models with only final answers makes learning difficult, since there are many possible reasoning paths that could lead to the same result. Without guidance, models may rely on shortcuts or dataset biases instead of learning the correct reasoning process. In contrast, providing intermediate annotations helps models identify the relevant steps needed to reach the answer, improving both performance and robustness (Dua et al., 2020). This perspective highlights why CoT prompting is effective: by generating intermediate reasoning steps at inference time, it provides a form of implicit supervision that guides the model toward more accurate and structured reasoning.

Eventually, CoT prompting has been shown to depend strongly on model scale. (Wei et al., 2022) observe that CoT reasoning emerges primarily in sufficiently large language models, while smaller models often generate reasoning steps that are fluent but logically incorrect. This suggests that CoT is not simply a prompting trick, but an ability that becomes effective only when models reach a certain level of capacity. As a result, CoT prompting has become a foundational method in LLM research, expanding the range of tasks that language models can solve and offering insight into how these models perform reasoning.

3 Mechanisms

3.1 Problem Decomposition

One reason Chain of Thought prompting improves performance is that it transforms complex reasoning tasks into a sequence of simpler decisions, effectively reducing the difficulty of the prediction problem. Rather than requiring the model to map an input directly to a final answer in a single step, CoT introduces intermediate reasoning states that guide the model toward the correct solution. This decomposition reduces the search space of possible outputs and allows the model to make locally coherent decisions at each step (Wei et al., 2022).

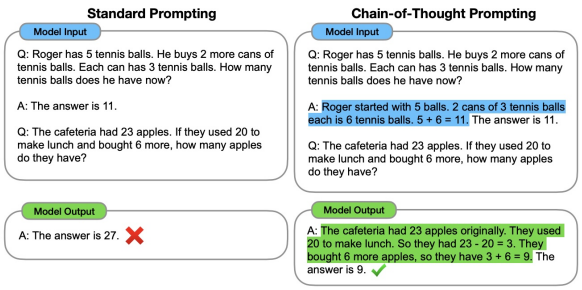

Figure 1: Illustration comparing standard prompting and chain-of-thought prompting. Chain-of-thought augments reasoning with intermediate steps, enabling improved performance on multi-step tasks (Wei et al., 2022).

This effect is illustrated in Figure 1, where standard prompting produces an incorrect answer by directly predicting the output, while CoT prompting generates intermediate reasoning steps that lead to the correct solution. The example highlights how step-by-step reasoning helps the model avoid errors that arise when solving the problem in a single step. By structuring the reasoning process, CoT enables more reliable handling of multi-step tasks such as arithmetic and logical reasoning (Wei et al., 2022).

3.2 Computation Through Natural Language Reasoning

CoT prompting improves performance by increasing the effective amount of computation applied during inference. Instead of producing a single output, the model generates a sequence of intermediate tokens that represent reasoning steps, allowing additional processing before arriving at a final answer. Importantly, studies demonstrate that this improvement is not simply a result of longer outputs. Ablation studies demonstrate that increasing token length without meaningful reasoning structure does not yield performance gains, indicating that the content and organization of intermediate steps are essential (Wei et al., 2022). This suggests that CoT enhances performance by directing how computation is used, rather than merely increasing its quantity.

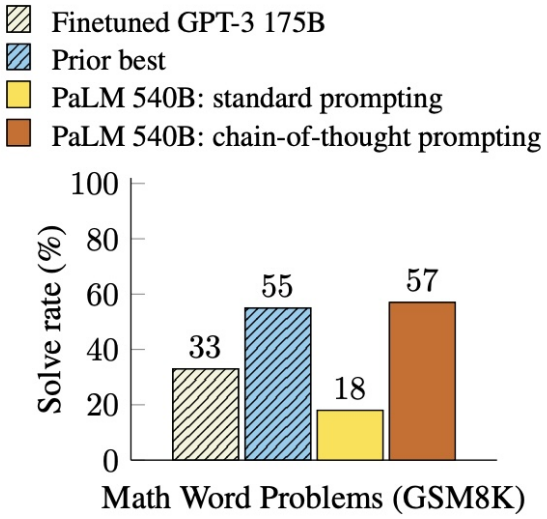

Figure 2: Chain-of-thought prompting significantly improves performance on GSM8K, enabling PaLM 540B to outperform standard prompting and prior methods (Wei et al., 2022; Cobbe et al., 2021).

The effect of this structured computation is reflected in experimental results. As shown in Figure 2, chain-of-thought prompting significantly improves performance on the GSM8K benchmark, increasing accuracy from 18% under standard prompting 57% when CoT is applied. This substantial gain indicates that CoT enables models to utilize their computational capacity more effectively when solving complex reasoning tasks. Notably, the largest improvements occur on problems that require multiple reasoning steps, reinforcing the idea that CoT benefits arise from structured reasoning processes rather than increased output length alone (Wei et al., 2022). This perspective is further supported by recent work extending chain-of-thought reasoning to multimodal settings. In vision-language models (VLMs), generating intermediate reasoning steps has been shown to improve both interpretability and generalization across diverse tasks, including visual question answering and chart understanding (Zhang et al., 2025). In contrast, training on short direct answers alone yields limited improvements in reasoning performance, while incorporating explicit chain-of-thought expla-

nations significantly enhances the model's ability to solve complex tasks. These results suggest that the effectiveness of CoT arises from a more general mechanism of structured intermediate computation, rather than being limited to text-based reasoning.

3.3 Structured Reasoning and Error Types

Despite its advantages, CoT prompting does not fully eliminate reasoning errors, which provides important insight into its limitations. Evidence identifies several common failure modes in CoT reasoning, including calculation errors, missing intermediate steps, and semantic misunderstandings (Wang et al., 2023a).

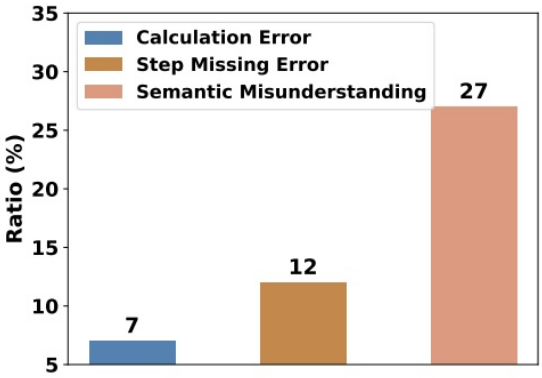

Figure 3: Distribution of error types in GSM8K solutions generated by zero-shot chain-of-thought prompting, highlighting calculation, missing-step, and semantic understanding errors (Wang et al., 2022; Wei et al., 2022).

As shown in Figure 3, analysis of incorrect predictions on the GSM8K benchmark reveals that semantic misunderstanding errors are the most frequent (27%), followed by missing-step errors (12%) and calculation errors (7%). This distribution indicates that many failures are not due to simple arithmetic mistakes, but rather to deeper issues in understanding the problem or organizing the reasoning process. To address these limitations, Plan-and-Solve prompting introduces an additional level of structure by explicitly separating reasoning into two stages: first generating a plan, and then executing that plan step by step. By requiring the model to identify relevant sub-tasks before solving them, this approach reduces missing-step errors and improves reasoning completeness. Furthermore, adding detailed instructions, such as extracting variables and verifying intermediate results, helps reduce calculation errors and improve overall reasoning quality (Wang et al., 2023a). These results show that CoT errors are primarily driven by reasoning quality rather than structure alone, motivating more structured approaches such as Plan-and-Solve to improve reliability

3.4 Prompt Quality and Sensitivity

Another important factor influencing the effectiveness of chain-of-thought prompting is its sensitivity to prompt design. Unlike the mechanisms discussed in previous sections, which focus on how reasoning is structured internally, CoT performance can vary significantly depending on how reasoning is elicited through input examples. In particular, different sets of exemplars, reasoning styles, and even annotators can lead to noticeable variation in performance, indicating that CoT is highly dependent on prompt construction rather than being a fully stable model capability (Wei et al., 2022).

This limitation is further explored by Active Prompting, which shows that standard CoT methods rely on a fixed set of manually designed examples that may not generalize well across tasks (Diao et al., 2024). Instead of treating all examples equally, Active Prompting selects examples based on model uncertainty, prioritizing those that the model finds most challenging. These high-uncertainty examples provide more informative reasoning patterns and lead to improved performance across multiple reasoning benchmarks (Diao et al., 2024).

These findings suggest that CoT reasoning is not solely determined by model size or internal structure, but is strongly influenced by how reasoning is triggered at inference time. As a result, CoT should be understood as a prompt-sensitive behavior rather than a fully reliable reasoning mechanism. Improving prompt selection and diversity is therefore critical for achieving consistent reasoning performance.

4 Accuracy and Reliability of CoT

Although chain-of-thought prompting often improves final-answer accuracy, the generated reasoning process is not always reliable or faithful to the model's true internal decision process. Results show that CoT can substantially improve benchmark performance, especially on multi-step reasoning tasks, but these gains do not guarantee that the intermediate reasoning itself is logically sound or free from bias (Wei et al., 2022; Wang et al., 2023a; Wu et al., 2024; Wang et al., 2023b). This section,

therefore, distinguishes between the accuracy gains associated with CoT and the reliability problems that remain even when performance improves.

4.1 Accuracy Improvements

CoT prompting is widely known to improve performance; a more important observation is how these improvements emerge. Rather than providing uniform gains across all settings, CoT exhibits a strong dependence on both model scale and task complexity. (Wei et al., 2022) show that CoT prompting yields minimal improvements for smaller models, but produces substantial accuracy gains once models reach sufficiently large scales.

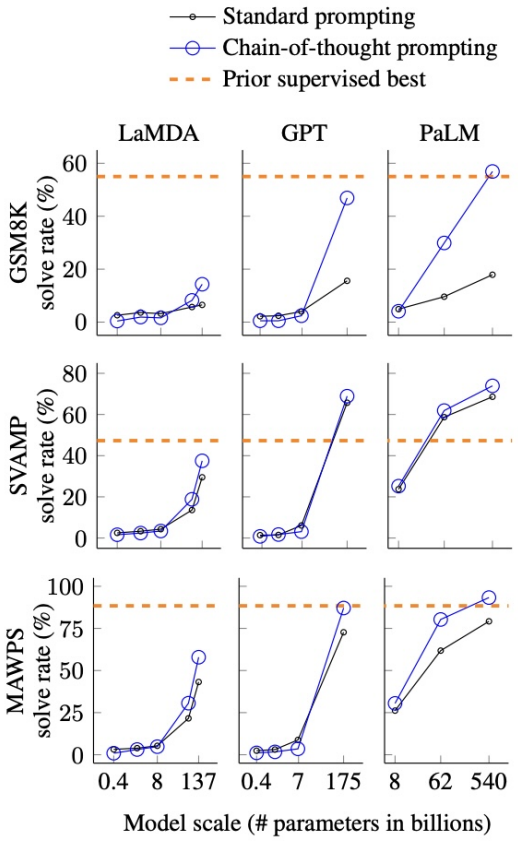

Figure 4: Chain-of-thought prompting improves mathematical reasoning performance and emerges with model scale across GSM8K, SVAMP, and MAWPS benchmarks (Wei et al., 2022; Cobbe et al., 2021; Jie et al., 2022; Lan et al., 2021).

This behavior is illustrated in Figure 4, which compares standard prompting and CoT prompting across multiple model families and reasoning benchmarks, including GSM8K, SVAMP, and MAWPS. The results show that for smaller models, performance differences between prompting strategies are negligible. However, as model size increases, CoT prompting begins to significantly outperform standard prompting, with large models such as PaLM 540B achieving substantial improvements in solve rates. This pattern suggests that CoT is not a universally effective technique, but rather an emergent capability that becomes usable only at larger scales (Wei et al., 2022).

In addition to scale, the benefits of CoT are also closely tied to task difficulty. The largest performance gains are observed on problems that require multiple reasoning steps, where direct prediction is insufficient. On simpler tasks, the advantage of CoT is much smaller, indicating that its primary benefit lies in enabling structured reasoning for complex problems rather than improving general performance across all tasks (Wei et al., 2022).

4.2 Bias, Faithfulness, and Incorrect Reasoning

Beyond structural errors, a more fundamental limitation of chain-of-thought prompting lies in the lack of faithfulness between generated reasoning and the model's actual decision process. Even when CoT produces fluent and detailed explanations, these reasoning chains are not always consistent with the model's predictions or grounded in correct logic. Findings indicate that language models can generate rationales that appear plausible but fail to justify the final answer, raising concerns about the interpretability of CoT reasoning (Wang et al., 2023a). This concern is further supported by prior work showing that generated rationales may be inconsistent with model predictions or fail to justify answers, often due to hallucinated or ungrounded reasoning (Wang et al., 2023b).

DeCoT provides a deeper explanation of this issue by showing that reasoning errors can be driven by biases in the model's internal knowledge. Specifically, the model may rely on irrelevant or spurious information when generating reasoning steps, leading to logically incorrect conclusions even when the reasoning appears coherent (Wu et al., 2024). This highlights a key limitation of CoT: generating more detailed reasoning does not necessarily eliminate bias, and may instead propagate incorrect assumptions through the reasoning chain. As a result, simply injecting external knowledge into prompts is not sufficient, since models may still fail to identify and use the relevant information correctly.

These findings suggest that CoT reasoning is not inherently faithful to the model's internal decision-

making process, and that correct answers do not always imply correct reasoning. In some cases, models may arrive at the correct answer through flawed or incomplete reasoning paths, while in others, they may produce highly structured but incorrect explanations. This disconnect between reasoning and prediction raises important challenges for the use of CoT in applications that require interpretability and trustworthiness.

Dataset	Question	CoTs w/ Context	CoTs w/o Context	CoT w/ Post-Editing	DeCoT
HotpotQA	The Pluralist school is said to have included what creator of the theory of atomism?	The Pluralist school was a school of pre-Socratic philosophers. The Pluralist school included Leucippus and Democritus. Therefore, the final answer is [Democritus ✗].	One of the key pluralist figures was Greek philosopher Aristotle. Aristotle is known for his theory that nature contains no vacuums. Therefore, the final answer is [Aristotle ✗].	One of the key pluralist figures was Greek philosopher Aristotle. Aristotle is known for his theory of atomism. Therefore, the final answer is [Aristotle ✗].	The Pluralist school is a philosophical movement that emerged in Ancient Greece. It is said to have included Leucippus, the creator of the theory of atomism. Therefore, the final answer is [Leucippus ✓].
MuSiQue	What county contains the city with a radio station that broadcasts to the capital city of the state where the Peace center is located?	The state where the Peace center is located is Greenville, South Carolina. The county containing the city with a radio station broadcasting to Greenville is likely Greenville County. Therefore, the final answer is [Greenville County ✗].	The Peace Center is located in Greenville, South Carolina. The capital city of South Carolina is Columbia. There are two radio stations in Greenville that broadcast to Columbia - WJMZ-FM and WMYI. WMYI, which is located in Greenville, broadcasts to the capital city of Columbia. Therefore, the final answer is [Greenville County ✗].	The Peace Center is located in Greenville, South Carolina. The capital city of South Carolina is Columbia. There are two radio stations in Greenville that broadcast to Columbia - WJMZ-FM and WMYI. WMYI is located in Hendersonville, North Carolina. Therefore, the final answer is [Hendersonville ✗].	The Peace Center is located in Greenville, South Carolina. The radio station WWNQ is licensed to Forest Acres and broadcasts to Columbia, which is the capital city of South Carolina. Forest Acres is a city in Richland County, South Carolina. Therefore, the final answer is [Richland County ✓].
SciQ	By exploding what the space probes get separated from their launchers?	The context is discussing how space probes use exploding bolts to separate from their launchers. The question is asking what the space probes are separated from during this process. Therefore, the final answer is [their launchers ✗].	When a space probe is launched, it is attached to a rocket or launcher. Once the rocket has reached a certain point in space, it explodes, releasing the space probe. Therefore, the final answer is [the rocket or launcher ✗].	When a space probe is launched, it is attached to a rocket or launcher. ~~Once the rocket has reached a certain point in space, it explodes, releasing the space probe.~~ Therefore, the final answer is [the explosion or detonation ✗].	The context states that space probes may be separated from their launchers by exploding bolts. Exploding bolts are bolts that explode or break apart in order to separate two objects. Therefore, the final answer is [bolts ✓].
WikiHop	What's the genre of Terra Brasilis?	"Genre" refers to a category of artistic work. Terra Brasilis is a famous album by Brazilian musician Milton Nascimento. Therefore, the final answer is [Brazilian music ✗].	"Terra Brasilis" is a song by Brazilian composer Caetano Veloso. The song is from his 1978 album "Muito (Dentro Da Estrela Azulada)". Therefore, the final answer is [Brazilian ✗].	"Terra Brasilis" is an album by Brazilian composer Antônio Carlos Jobim. ~~The song is from his 1978 album "Muito (Dentro Da Estrela Azulada)".~~ Therefore, the final answer is [Brazilian ✗].	The album Terra Brasilis was released in 1980 by Brazilian composer Antônio Carlos Jobim. Jobim is known for his contributions to bossa nova music. Therefore, the final answer is [bossa nova ✓].

Table 1: Comparison of chain-of-thought variants highlighting factual and logical errors versus corrected reasoning; DeCoT improves reasoning accuracy by mitigating bias in knowledge-intensive tasks (Wu et al., 2024).

Illustrative examples of these limitations are shown in Table 1, which compares reasoning outputs generated by standard CoT prompting, CoT without context, and DeCoT. The examples show that standard CoT often produces reasoning chains containing factual inaccuracies and logical errors, even when the reasoning appears coherent. In contrast, DeCoT is able to generate reasoning that better aligns with relevant evidence, leading to correct final answers. These results highlight that incorrect reasoning in CoT is not simply due to missing steps, but often arises from biased or misleading intermediate reasoning paths, reinforcing the need for methods that explicitly address reasoning faithfulness (Wu et al., 2024).

Finally, while prompt design plays a central role in eliciting reasoning, the generation process itself can also influence reasoning quality. Since CoT relies on autoregressive token generation, the selection of intermediate tokens can affect the coherence and correctness of reasoning paths. This suggests that reasoning performance depends not only on prompt structure, but also on how reasoning trajectories are generated and explored during inference. Together, these limitations highlight the need for methods that improve both the structure and faithfulness of reasoning in large language models.

5 Variants and Comparison of Prompting Strategies

Recent work has explored a range of prompting strategies that improve chain-of-thought reasoning through different mechanisms, including structured decomposition, example selection, and reasoning path exploration (Wang et al., 2023a; Zhou et al., 2023; Diao et al., 2024).

5.1 Standard Prompting

Standard prompting serves as the baseline inference paradigm for large language models, where the model is expected to map an input directly to an output without generating intermediate reasoning steps. Rather than explicitly structuring the reasoning process, the model relies on patterns learned during pretraining and in-context examples to produce a final answer in a single forward pass (Wei et al., 2022).

From a computational perspective, standard prompting can be understood as performing implicit reasoning within the model's hidden representations, without exposing intermediate steps in natural language. While this allows for efficient inference, it also places a greater burden on the model to internally resolve complex dependencies in a single generation trajectory. As a result, performance is highly sensitive to the model's ability to compress multi-step reasoning into a direct prediction.

Research has shown that this direct prediction paradigm often leads to unstable behavior on tasks requiring compositional reasoning, where multiple intermediate decisions must be made sequentially. In such settings, standard prompting tends to produce answers that reflect shortcut heuristics or dataset biases, rather than a structured reasoning process (Wei et al., 2022). This limitation is not merely a lack of explanation, but a consequence of how inference is performed, as the model is not explicitly guided to decompose the problem.

Despite these limitations, standard prompting remains an important reference point for evaluating reasoning methods. Many improvements observed in chain-of-thought and related approaches can be understood as modifying or extending this baseline by introducing intermediate reasoning steps, restructuring the inference process, or exploring multiple reasoning trajectories. Therefore, standard prompting provides a useful foundation for understanding how different prompting strategies

alter the behavior of large language models.

5.2 Chain-of-Thought Prompting

CoT prompting extends the standard prompting paradigm by explicitly guiding the model to generate intermediate reasoning steps before producing a final answer. Instead of relying on a single-step prediction, the model is encouraged to follow a structured reasoning trajectory, where each step incrementally contributes to the solution (Wei et al., 2022). This shift transforms the inference process from implicit reasoning within hidden representations to an explicit sequence of natural language reasoning steps.

From a comparative perspective, the key advantage of CoT prompting lies in its ability to improve performance on tasks that require multi-step reasoning. Additionally, CoT reduces the difficulty of generating correct answers in a single pass and enables the model to handle problems that involve arithmetic operations, logical dependencies, or compositional reasoning (Wei et al., 2022). As a result, CoT prompting has been shown to significantly outperform standard prompting on reasoning benchmarks, particularly when applied to sufficiently large models.

However, CoT prompting should not be understood as a complete solution to reasoning. While it introduces a more structured inference process, the quality of intermediate reasoning steps can vary, which may affect the reliability of final predictions (Wang et al., 2023a). This limitation motivates the development of more advanced prompting strategies that build upon CoT, such as structured planning methods and ensemble-based approaches (Zhou et al., 2023; Wang et al., 2023b). Within the broader landscape of prompting techniques, CoT serves as a foundational method that bridges the gap between direct prediction and more structured reasoning frameworks. Many subsequent approaches can be viewed as extensions of CoT that aim to improve the organization, reliability, or selection of reasoning steps during inference.

5.3 Structured Reasoning Approach: Plan-and-Solve

Structured reasoning approaches extend chain-of-thought prompting by explicitly controlling how reasoning is organized before solution generation. Instead of relying on the model to implicitly discover a sequence of reasoning steps, these methods guide the model to first construct a high-level plan and then execute it in a structured manner. In particular, Plan-and-Solve prompting introduces a two-stage reasoning process in which the model first decomposes the task into subproblems and then solves each component step by step (Wang et al., 2023a). This approach shifts reasoning from a reactive process into a more deliberate and organized procedure.

A key advantage of this formulation is that it reduces reliance on the model's implicit reasoning abilities. In standard CoT prompting, the model generates reasoning in a single pass, which can lead to inconsistent or poorly structured solution paths. In contrast, Plan-and-Solve enforces an explicit decomposition of the task before execution, ensuring that the reasoning process follows a predefined structure. This improves the coherence of intermediate steps and helps align the reasoning process with the logical requirements of the task (Wang et al., 2023a).

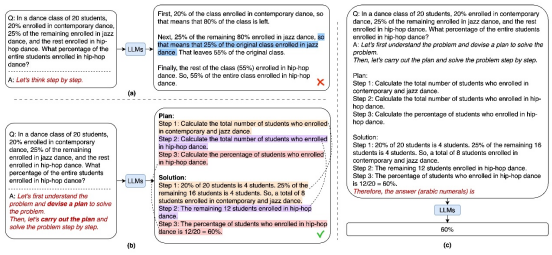

Figure 5: Comparison between Zero-shot-CoT and Plan-and-Solve prompting. While Zero-shot-CoT generates reasoning directly, Plan-and-Solve first constructs a plan and then executes it step by step, resulting in more structured and accurate reasoning (Wang et al., 2023a).

As shown in Figure 5, Zero-shot-CoT generates reasoning by directly following prompts such as "Let's think step by step," which can result in incomplete or inconsistent reasoning when the problem requires careful decomposition. In contrast, Plan-and-Solve explicitly separates reasoning into a planning phase and a solution phase. The example demonstrates that by first identifying the necessary sub-steps and then executing them sequentially, the model produces a more coherent and logically consistent reasoning chain. This highlights that the improvement of Plan-and-Solve does not come from simply adding more reasoning steps, but from structuring the reasoning process before it begins, allowing the model to better organize and execute complex problem-solving tasks.

5.4 Least-to-Most Prompting

Least-to-most prompting extends structured reasoning by explicitly decomposing complex problems into a sequence of simpler subproblems that are solved incrementally. Unlike standard chain-of-thought prompting, where reasoning is generated in a single pass, this approach separates reasoning into two sequential stages: (1) problem decomposition and (2) subproblem solving (Zhou et al., 2023). In the first stage, the model is prompted to break down the original problem into intermediate questions. In the second stage, the model solves each subproblem step by step, using previously generated answers to guide subsequent reasoning.

This two-stage design enables progressive reasoning, where each step builds directly on prior outputs. As described by (Zhou et al., 2023), solving each subproblem is "facilitated by the answers to previously solved subproblems," allowing the model to construct solutions in a compositional manner rather than generating reasoning from scratch. This differs from standard CoT prompting, where intermediate steps are generated independently within a single reasoning chain, without explicitly reusing prior partial solutions.

A key advantage of least-to-most prompting is its ability to generalize to problems that are more complex than those seen in the prompt examples. Experimental results show that while chain-of-thought prompting struggles with tasks requiring longer reasoning chains, least-to-most prompting maintains strong performance by decomposing problems into manageable components (Zhou et al., 2023). This highlights that structuring reasoning as a sequence of dependent subproblems can improve both scalability and robustness in multi-step reasoning tasks.

Figure 6 illustrates how least-to-most prompting enforces a dependency between intermediate steps, where each subproblem builds directly on the solution of the previous one. This sequential structure allows the model to reuse partial results, leading to more coherent and complete reasoning compared to single-pass generation. Overall, least-to-most prompting demonstrates that effective reasoning in large language models depends not only on generating intermediate steps but also on organizing those steps into a coherent and reusable structure. By enforcing decomposition and sequential dependency, this approach provides a more controlled alternative to standard CoT prompting, particularly for tasks that require deeper compositional reasoning.

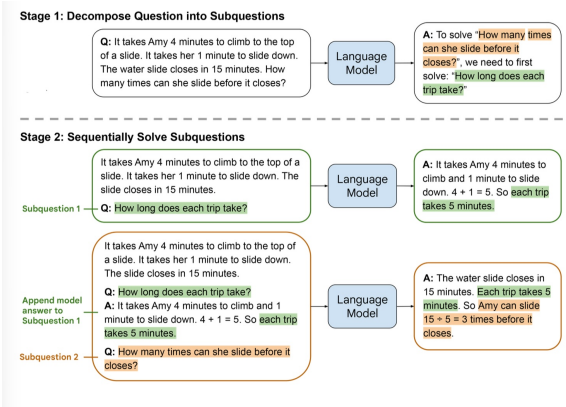

Figure 6: Illustration of least-to-most prompting, where complex problems are decomposed into subproblems and solved sequentially, with later steps conditioned on earlier answers (Zhou et al., 2023).

5.5 Active Prompting and Example Selection

Active Prompting reframes chain-of-thought prompting as a problem of example selection rather than only reasoning format. Standard CoT methods assume that a small fixed set of human-written exemplars is sufficient to elicit strong reasoning, but this assumption is often too rigid for tasks that vary in difficulty, scope, and domain. (Diao et al., 2024) argue that manually chosen exemplars are "not necessarily the most effective" for different reasoning tasks, even when they follow the correct chain-of-thought format. From this perspective, prompt optimization is not simply about adding more examples, but about selecting examples that are most informative for the target task.

To address this limitation, Active Prompting introduces an uncertainty-based selection strategy for choosing which questions should be annotated with CoT rationales. Instead of relying on a fixed prompt, the method queries the model multiple times on candidate inputs, measures uncertainty based on the diversity of generated answers, and selects the most uncertain instances for annotation (Diao et al., 2024). The intuition is that uncertain or difficult examples expose gaps in the model's reasoning ability, making them more valuable for constructing effective prompts.

A key contribution of this approach is its connection to active learning. (Diao et al., 2024) show that disagreement and entropy-based uncertainty metrics are effective for selecting informative examples, while self-confidence is unreliable due to the tendency of language models to be overconfident. This shifts prompt design from a static pro-

cess into an adaptive one, where exemplar selection is guided by the model's own behavior rather than fixed human intuition.

METHOD	GSM8K	ASDIV	SVAMP	AQUA	SINGLEEQ	CSQA	STRATEGY	LETTER (4)	AVG.
Prior Best	55.0[a]	75.3[b]	57.4[c]	37.9[d]	32.5[e]	91.2[f]	73.9[g]	-	-
text-davinci-002									
Auto-CoT	47.9	-	69.5	36.5	87.0	74.4	65.4	59.7	-
CoT	46.9	71.3	68.9	35.8	77.3	73.5	65.4	56.6	61.5
SC	58.2	76.9	78.2	41.8	87.2	72.9	70.7	57.6	67.9
Random-CoT	63.9	82.3	81.1	44.1	89.4	74.5	73.3	65.5	71.8
Active-Prompt (D)	**73.2**	83.2	**82.7**	48.4	90.6	76.6	**76.9**	**67.7**	74.9
Active-Prompt (E)	71.1	**83.8**	81.8	**50.3**	**93.1**	**78.8**	**76.9**	66.7	**75.3**
code-davinci-002									
Auto-CoT	62.8	-	-	-	-	-	-	-	-
CoT	63.1	80.4	76.4	45.3	93.1	77.9	73.2	70.4	72.5
SC	78.0	87.8	86.8	52.0	93.7	81.5	79.8	73.4	79.1
Random-CoT	78.6	87.1	88.0	53.1	94.0	82.1	79.4	73.3	79.4
Active-Prompt (D)	82.2	88.4	**88.7**	55.1	94.5	**83.9**	**80.6**	74.1	80.9
Active-Prompt (E)	**83.4**	**89.3**	87.5	**57.0**	**95.5**	82.6	**80.6**	**76.7**	**81.6**
gpt-3.5-turbo-0613 (w.o. SC)									
CoT	74.2	82.5	83.8	50.0	95.0	79.9	80.5	82.0	78.5
Active-Prompt (D)	77.1	83.6	85.5	50.0	**96.0**	**81.5**	**82.1**	**84.0**	80.0
Active-Prompt (E)	**78.2**	**84.7**	**86.0**	**57.3**	95.5	80.7	81.3	**84.0**	**81.0**
gpt-3.5-turbo-0301 (w.o. SC)									
CoT	80.1	86.7	82.0	56.2	91.3	74.6	64.4	81.4	77.1
Active-Prompt (D)	83.5	87.4	83.0	60.6	93.3	**75.9**	70.0	**84.0**	79.7
Active-Prompt (E)	**83.8**	**88.8**	**83.7**	**61.0**	**93.7**	75.0	**71.0**	**84.0**	**80.1**

Table 2: Performance comparison of Active Prompting with CoT, Self-Consistency (SC), and Random CoT across multiple reasoning benchmarks (adapted from (Diao et al., 2024)).

As shown in Table 2, Active Prompting consistently outperforms standard CoT, self-consistency, and random exemplar selection across a wide range of reasoning tasks, including arithmetic (GSM8K), commonsense (CSQA), and symbolic reasoning benchmarks. The results demonstrate that Active Prompt variants achieve higher average performance across different model architectures, including text-davinci-002 and GPT-3.5-based models. Notably, the improvements are not limited to a single task, but are observed across multiple datasets, indicating that uncertainty-based exemplar selection provides a more general mechanism for improving reasoning performance. These findings suggest that the effectiveness of CoT prompting depends not only on generating intermediate reasoning steps but also on selecting examples that better reflect challenging reasoning patterns (Diao et al., 2024).

Within the landscape of prompting strategies, Active Prompting represents a shift from static prompt design to adaptive prompt construction. Rather than treating exemplars as fixed inputs, it incorporates a feedback loop in which the model's uncertainty guides the selection of informative examples. This highlights that improving reasoning in large language models is not only a matter of how reasoning is structured, but also of how it is elicited through carefully chosen prompts.

5.6 Ensemble and Self-Consistency Methods

Ensemble-based reasoning strategies extend CoT by generating multiple reasoning paths and aggregating them into a final answer. The central approach, self-consistency (SC), samples multiple CoT outputs and selects the most frequent answer through majority voting (Wang et al., 2023b). Because incorrect reasoning paths tend to produce inconsistent or low-frequency answers, aggregation acts as a filtering mechanism that reduces the impact of individual errors. However, performance typically scales with the number of sampled paths, making SC computationally expensive (Wan et al., 2025).

Improving this efficiency-accuracy trade-off has motivated recent work. SCOTT addresses faithfulness by aligning generated rationales with final predictions through counterfactual objectives, reducing reliance on spurious reasoning (Wang et al., 2023b). The Reasoning-Aware Self-Consistency (RASC) framework extends SC by scoring samples based on reasoning coherence rather than treating all outputs equally, enabling weighted voting and adaptive early stopping (Wan et al., 2025).

Model	Method	Benchmark Category								
		Mathematical Reasoning			Commonsense Reasoning			Symbolic Reasoning		
		# of Gen	Acc	Gain/Sample	# of Gen	Acc	Gain/Sample	# of Gen	Acc	Gain/Sample
GPT-4	CoT	1	82.1%	-	1	83.2%	-	1	92.5%	-
	SC	40	87.5%	0.139%	40	88.0%	0.123%	40	97.3%	0.123%
	ESC	6.86 (-82.9%)	87.3%	0.889%	7.93 (-80.2%)	88.5%	0.767%	5.54 (-86.2%)	97.3%	1.056%
	AC	5.89 (-85.3%)	87.3%	1.065%	6.15 (-84.6%)	87.2%	0.779%	4.43 (-88.9%)	97.3%	1.395%
	RASC	**4.59** (-88.5%)	87.5%	**1.503%**	**4.74** (-88.1%)	88.3%	**1.367%**	**4.19** (-89.5%)	97.3%	**1.500%**
GPT-3.5 Turbo	CoT	1	63.5%	-	1	71.2%	-	1	80.1%	-
	SC	40	69.1%	0.144%	40	76.1%	0.126%	40	85.3%	0.134%
	ESC	14.94 (-62.7%)	69.3%	0.417%	10.86 (-72.9%)	76.4%	0.527%	7.55 (-81.1%)	86.0%	0.903%
	AC	12.49 (-68.8%)	67.6%	0.357%	8.34 (-79.2%)	75.0%	0.517%	6.39 (-84.0%)	84.7%	0.853%
	RASC	**6.62** (-83.5%)	69.4%	**1.054%**	**4.95** (-87.6%)	76.1%	**1.238%**	**4.36** (-89.1%)	85.3%	**1.547%**
Vicuna-13B	CoT	1	36.2%	-	1	50.1%	-	1	41.5%	-
	SC	40	41.7%	0.141%	40	55.9%	0.149%	40	47.3%	0.149%
	ESC	20.29 (-49.3%)	40.5%	0.223%	20.89 (-47.8%)	50.5%	0.020%	23.05 (-42.4%)	44.0%	0.113%
	AC	17.44 (-56.4%)	42.3%	0.371%	19.92 (-50.2%)	55.9%	0.306%	25.77 (-35.6%)	49.3%	0.314%
	RASC	**8.24** (-79.4%)	42.0%	**0.801%**	**7.83** (-80.4%)	55.0%	**0.718%**	**8.71** (-78.2%)	45.3%	**0.494%**
Llama2-7B	CoT	1	18.5%	-	1	63.2%	-	1	19.5%	-
	SC	40	23.0%	0.115%	40	68.9%	0.146%	40	24.2%	0.121%
	ESC	27.25 (-31.9%)	23.5%	0.190%	8.85 (-77.9%)	68.9%	0.726%	31.03 (-22.4%)	24.8%	0.177%
	AC	23.91 (-40.2%)	21.5%	0.131%	7.97 (-80.1%)	68.3%	0.735%	24.62 (-38.5%)	24.2%	0.199%
	RASC	**10.71** (-73.2%)	23.2%	**0.484%**	**5.11** (-87.2%)	68.9%	**1.390%**	**13.09** (-67.3%)	24.8%	**0.439%**

Table 3: Performance of self-consistency variants across mathematical, commonsense, and symbolic reasoning benchmarks. Methods include Chain-of-Thought (CoT), Self-Consistency (SC), Early-Stopping Self-Consistency (ESC), Adaptive Consistency (AC), and Reasoning-Aware Self-Consistency (RASC). Metrics report the number of generated samples, accuracy, and accuracy gain per sample relative to CoT (Wan et al., 2025).

The effectiveness of these approaches is illustrated in Table 3, where multiple self-consistency variants are compared across mathematical, commonsense, and symbolic reasoning benchmarks. Standard SC improves accuracy over single-path CoT but requires a large number of samples (e.g., 40 generations). In contrast, adaptive methods such as ESC and AC reduce the number of samples while maintaining similar accuracy. Notably,

RASC achieves the best trade-off, reducing sampling by up to approximately 70–90 percent while preserving or slightly improving accuracy, and significantly increasing accuracy gain per sample (Wan et al., 2025). This demonstrates that incorporating reasoning quality into the aggregation process leads to more efficient and reliable inference.

6 Inference-Time Reasoning and Demonstration Construction

CoT reasoning operates through sequential token generation, (Feng et al., 2023) shows that this process increases the effective computational depth of language models, enabling them to solve problems that are difficult for single-step prediction.

Its effectiveness also depends on how reasoning demonstrations are constructed. Auto-CoT shows that diverse, automatically generated examples can replace manual prompts and improve robustness by reducing reliance on imperfect reasoning chains (Zhang et al., 2022).

Overall, CoT performance depends not only on reasoning format, but also on the interaction between generation dynamics and demonstration design.

7 Experiment Set Up and Results

7.1 Experimental Set Up

To complement the literature review, I implemented a small prompting experiment comparing three inference strategies: standard prompting, chain-of-thought prompting, and chain-of-thought with self-consistency. The experiment uses a locally served Mistral model through an OpenAI-compatible API provided by Ollama, which enables running large language models on local hardware (Ollama, 2024). Each method is evaluated on the same set of manually verifiable reasoning questions covering arithmetic, symbolic reasoning, multi-step reasoning, rate reasoning, division, and comparison tasks.

The goal of the experiment is not to reproduce large-scale benchmark results, but to provide a controlled comparison of how prompting strategy affects reasoning behavior on short problems with known answers. Standard prompting asks the model to provide only a final answer, CoT prompting asks the model to reason step by step before answering, and self-consistency samples multiple CoT responses before selecting the most frequent final answer.

7.2 Evaluation Metrics

Model outputs are evaluated using exact answer correctness after answer normalization. Because LLMs do not always follow the requested output format, the evaluation script extracts final answers using both explicit "Final answer:" markers and fallback rules for numerical and yes/no outputs. Incorrect responses are also categorized into simple error types, including calculation errors, reasoning errors, formatting errors, missing answers, and other errors.

7.3 Results

Method	Initial Accuracy	Trial Accuracy
Standard Prompting	0.00	0.50
CoT Prompting	1.00	0.92
CoT + Self-Consistency	1.00	0.92

Table 4: Accuracy comparison across prompting methods for the initial run and repeated trials.

Table 4 summarizes the accuracy across prompting methods for both the initial run and repeated trials. While the initial run shows perfect performance for CoT-based methods, this result does not generalize across trials, where accuracy stabilizes at a lower but consistent level.

Across repeated trials, both CoT and CoT with self-consistency outperform standard prompting. This confirms that explicit reasoning improves performance over direct prediction. However, the difference between CoT and self-consistency is limited in this small-scale setting.

Run	Method	Calc.	Reason.	Format/Other
Initial	Standard	5	0	0
Initial	CoT	0	0	0
Initial	CoT + SC	0	0	0
Trials	Standard	5	0	1
Trials	CoT	0	0	1
Trials	CoT + SC	0	1	0

Table 5: Error type comparison across prompting methods for the initial run and repeated trials. Format and other errors are grouped for simplicity.

Table 5 shows that standard prompting mainly fails through calculation errors, while CoT-based methods remove calculation errors almost entirely. The remaining CoT errors are isolated reasoning or output inconsistencies.

8 Conclusion

This paper examined how chain-of-thought prompting improves reasoning in large language models through both theoretical analysis and a small-scale experiment. Prior sections show that CoT enhances performance by restructuring the inference process, enabling models to decompose complex problems into intermediate steps and allocate computation more effectively.

These findings are supported by experimental results, where CoT-based methods consistently outperform standard prompting, achieving perfect accuracy in the initial run (1.00) and stabilizing at 0.92 across repeated trials, compared to standard prompting, which improves from 0.00 to 0.50. This improvement is accompanied by a clear reduction in calculation errors, indicating that structured reasoning leads to more reliable multi-step problem solving.

However, both the literature and experimental results highlight that improved accuracy does not guarantee faithful reasoning. Generated reasoning chains may still be inconsistent or misleading, and the benefits of self-consistency remain limited in small-scale settings. Overall, CoT improves performance by organizing computation rather than fundamentally changing reasoning ability, demonstrating that structured inference plays a central role in enabling effective reasoning in large language models.

Limitations and Future Work

Despite its effectiveness, CoT prompting is not universally reliable. Recent work shows that in pattern-based in-context learning tasks, CoT and related reasoning strategies can underperform direct answering, with performance degrading as the number of demonstrations increases (Zheng et al., 2025). This limitation is attributed to the additional reasoning steps introducing contextual overhead, which interferes with pattern extraction and weakens implicit reasoning.

Another key limitation is the lack of reasoning faithfulness. Even when CoT improves final-answer accuracy, generated reasoning chains may be inconsistent, misleading, or disconnected from the model's actual decision process (Lanham et al., 2023; Wu et al., 2024). This raises concerns about interpretability and reliability in practical applications.

Future work should focus on developing adaptive reasoning strategies that balance explicit and implicit reasoning, rather than relying solely on step-by-step explanations. Approaches that improve reasoning faithfulness, reduce unnecessary reasoning overhead, and dynamically adjust prompting based on task structure may lead to more robust and generalizable reasoning systems.

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